

Elijah Croft

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EDUCATION

California State University Fullerton

Fullerton, CA

Double Major: B.S. Computer Science, B.A. Philosophy

Anticipated Graduation: May 2028

- Research: AI Ethics; Presented panel on algorithmic decision-making.

EXPERIENCE

Software Engineer (Contract)

May. 2025 – Present

2E4ME

Newport Beach, CA

- Designed and deployed a full-stack scheduling platform serving 20+ instructors and 70+ students, streamlining timetable management and reducing manual scheduling effort.
- Implemented a constraint-based optimization engine using OptaPlanner to automatically generate timetables.
- Designed subdomain infrastructure with Google OAuth 2.0 SSO, enabling secure internal authentication.

Robotics Coach/Programming Instructor

Mar. 2025 - Present

Magikid Robotics Lab

Tustin, CA

- Led a team of 10 to the VEX AI World Championships, achieving 18th and 8th place among international teams.
- Lead team to winning the VEX AI robotics international open at UC Berkeley.
- Assisted with AI integration in youth robotics education by implementing autonomous navigation based on the Markov Decision Process, reducing path planning errors by 80% in competition scenarios, in addition to CV via yolo11

PROJECTS

MixSafe | *Next.js, TypeScript, Tailwind CSS, Django* | **Demo** | **GitHub**

Oct. 2024 - Present

- Built and deployed a harm-reduction platform serving 500+ weekly users and processing 2,000 monthly interactions.
- Developed a structured interaction database enabling fast, real-time lookups for substance combinations and effects.

PPP Simulator (Best Use of ArcGIS) | *Next.js, TypeScript, ArcGIS SDK, Leaflet* | **Demo** | **GitHub**

Apr. 2026

- Developed a browser-based ocean pollution prediction platform that models the movement of plastic, oil, and chemical contaminants using current-field vector simulation and geospatial visualization.
- Engineered the core drift-simulation engine by processing real ArcGIS ocean-current data, applying vector-field calculations to predict pollutant trajectories and visualize long-term environmental impact.
- Integrated ArcGIS Maps SDK and Leaflet to render interactive marine hazard overlays/ocean-current visualizations.

C++ Chess Engine | *C++, Makefile, Git, Bitboards*

Jan. 2026 – Present

- Designed and developed a high-performance chess engine in C++ using bitboards and bitwise operations for efficient board representation and move generation.
- Implemented alpha-beta pruning, move ordering, and transposition tables to optimize evaluation and search.
- Developed adjustable engine-strength controls and complete legal move generation, including castling and en passant.

Automated Delivery Robot | *C++, Python, ROS2, Raspberry Pi 5, LiDAR, CAD*

Aug. 2025

- Engineered an autonomous ROS2-based robot featuring LiDAR localization, navigation, and obstacle avoidance.
- Designed power system with a 12V rechargeable ion battery, BMS, and motor driver; built aluminum chassis

VTO Alert Bot | *Python, Playwright, GraphQL, Raspberry Pi, Linux*

2026

- Developed a Python automation tool that monitors Amazon A to Z's live GraphQL endpoint for eligible Voluntary Time Off opportunities and sends real-time mobile notifications through ntfy.
- Deployed the bot as a persistent systemd service on a Raspberry Pi 4, using Chromium, Xvfb, and secure SSH-tunneled VNC to maintain an authenticated browser session without a physical display.
- Implemented configurable 20–30 second polling, pause and resume controls, login-expiration alerts, and secure environment-based credential management; validated the full detection and notification workflow against a live VTO drop.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript/TypeScript, Java, Scala

Tools: Git, Docker, Linux, Bash

Embedded: ROS2, Raspberry Pi, Jetson Orin Nano, LiDAR, Microcontrollers